METRIC
MIL-PRF-49291/6D
03 July 2018
SUPERSEDING
MIL-PRF-49291/6C
10 January 2003

PERFORMANCE SPECIFICATION SHEET

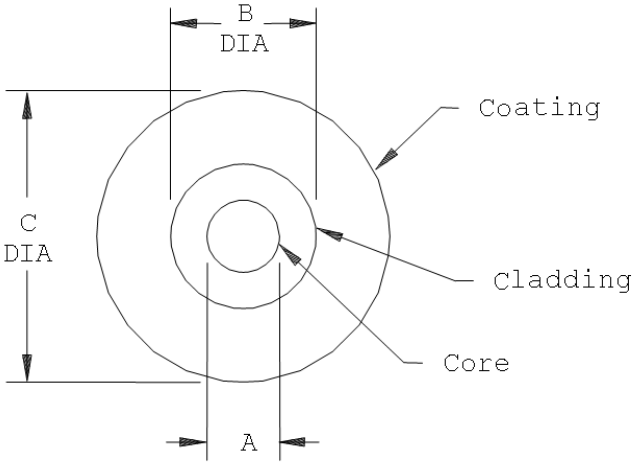
FIBER, OPTICAL, TYPE I, CLASS 1, SIZE IV, COMPOSITION A, WAVELENGTH B, RADIATION RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-49291.

DIMENSIONS AND CONFIGURATION REQUIREMENTS:

Diameter: See figures 1 and 2. (Diameter requirements are based on selecting fibers with end-point tolerances of 125±1 micrometer [µm] from production with a 125±2-µm tolerance.)



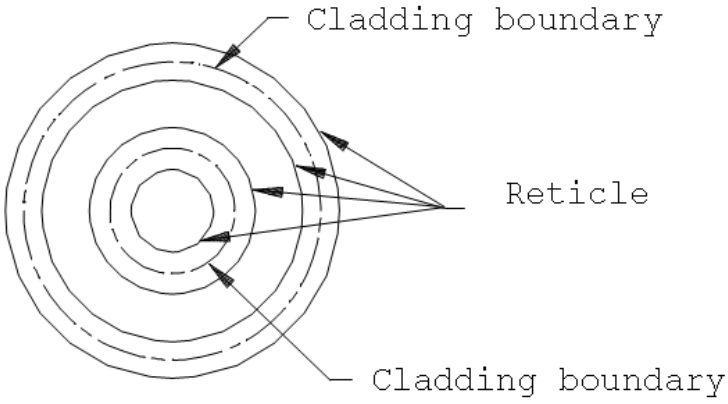
Fiber	PIN	Dimensions		
		A (µm)	B (µm)	C (µm)
Space qualified	M49291/6-02S	62.5±3	125±1	250±15
Standard	M49291/6-03	62.5±3	125±1	250±15
Space qualified	M49291/6-04S	62.5±3	125±1	500±25
Standard	M49291/6-05	62.5±3	125±1	500±25
Standard-blown fiber 1/	M49291/6-05BF	62.5±3	125±1	500±25

NOTE:
1/ Intended for use in blown fiber applications with MIL-PRF-85045 tube cable.

FIGURE 1. Dimensions and configuration of optical fiber construction.

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Circle (solid)	Diameter (µm)
Inner	59.5
Second	65.5
Third	124.0
Fourth	126.0

FIGURE 2. Tolerance fields.

Orality:

Core: ≤ 6 percent.

Cladding: ≤ 2 percent.

Offset:

Core-to-cladding: ≤ 4 µm.

Fiber-to-coating: Coating-cladding concentricity error ≤ 10.5 µm.

Overall coating concentricity ratio (OCCR) ≥ 0.70 for 250-µm diameter coatings and ≥ 0.84 for 500-µm diameter coatings.

Splices: Not allowed.

Tensile strength (proof test): 690 megapascals (MPa).

Fiber mass/unit length: 0.1 kilogram/kilometer (kg/km) maximum.

0.23 kg/km maximum for M49291/6-05BF.

Change in optical transmittance: Measurements to be made at 1300±20 nanometers (nm).

Maximum attenuation rate: 3.5 decibels/kilometer (dB/km) at 850±20 nm.

1.0 dB/km at 1300±20 nm.

Numerical aperture: 0.275±0.015 at 850±25 nm.

Overfill bandwidth: ≥ 600 megahertz-kilometers (MHz-km) at 1300±20 nm.

≥ 300 MHz-km at 850±20 nm.

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RML/EMBc bandwidth: ≥ 700 MHz-km at 1300±20 nm.

≥ 385 MHz-km at 850±20 nm.

Chromatic dispersion: The zero dispersion wavelength shall be not less than 1320 nm and not greater than 1365 nm. The dispersion slope at the zero dispersion wavelength shall be not greater than 0.11 picosecond/(nanometer^2-kilometer) [ps/(nm2·km)].

Transient attenuation: ≤ 1.5 dB at 1300 nm.

Macrobend attenuation: Performed at 1300±20 nm.

ENVIRONMENTAL REQUIREMENTS:

Temperature range: See table I.

TABLE I. Temperature range.

PIN	Operating (°C)	Nonoperating (°C)	Storage (°C)
M49291/6-02S M49291/6-04S	-55 to +85	-62 to +85	-62 to +85
M49291/6-03 M49291/6-05	-46 to +85	-55 to +85	-55 to +85
M49291/6-05BF	-28 to +85	-55 to +85	-55 to +85

Fluid immersion aging: Not applicable.

Dynamic tensile strength: Applicable.

Fungus: Applicable.

Nuclear radiation resistance: Nuclear radiation resistance requirements and test conditions shall be as shown below in tables II and III.

TABLE II. Steady state gamma radiation test conditions.

PIN	Test temperature (°C)	Dose rate	Total dose
M49291/6-02S M49291/6-04S	25±2 85±2	50 rad (Si)/minute	10 krad (Si)
M49291/6-03 M49291/6-05 M49291/6-05BF	-28±2 25±2 85±2	50+0, -20 rad (Si)/second	Classified 1/

NOTE:

1/ Some nuclear radiation resistance characteristics of this optical fiber are classified. Applications to receive these requirements can be made through the Department of the Navy, Naval Surface Warfare Center, Dahlgren Division, ATTN: Code A31, 17320 Dahlgren Road, Dahlgren, VA 22448-5100. Information concerning security clearance classification and "need to know" shall be detailed in the request.

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TABLE III. Steady state gamma radiation test requirements.

PIN	Maximum induced attenuation (dB/km)	Attenuation at specified recovery time (dB/km)	Specified recovery time (seconds)
M49291/6-02S M49291/6-04S	≤ 6	≤ 5	10,000
M49291/6-03 M49291/6-05 M49291/6-05BF	≤ 50 1/	≤ 15 at -28 °C 1/ ≤ 5 at 25 °C ≤ 5 at 85 °C	1,000
NOTE: 1/ The radiation induced loss for a given threat. The total dose associated with the threat is classified and not necessarily equal to the test total dose.			

Light launch conditions: In accordance with TIA-455-78.

Wavelength: 1300±25 nm.

Source type: Light-emitting diode (LED) with full width at half maximum (FWHM) spectral width ≤ 160 nm.

CONFORMANCE:

Qualification: Qualification shall consist of the testing specified in table IV.

TABLE IV. Qualification inspection.

Test	Initial qualification
Group I (three spools - 2 km each) 1/	
Visual and mechanical	X
Optical Time Domain Reflectometer (OTDR) tests:	
Fiber length	X
Attenuation uniformity	X
Parameter tests:	
Attenuation rate	X
Numerical aperture (MM only)	X
Core diameter (MM only)	X
Transient attenuation (MM only)	X
Macrobend attenuation	X
Bandwidth-overfill (MM only)	X
Bandwidth-RML/EMBc (MM only)	X
Chromatic dispersion	X
Mass/unit length	X
Geometry tests:	
End view:	
Core ovality (MM only)	X

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TABLE IV. Qualification inspection - Continued.

Test	Initial qualification
Cladding diameter	X
Cladding ovality	X
Core-to-cladding offset	X
Side view [2/]:	
Coating diameter	X
Coating-cladding concentricity error	X
Overall coating concentricity ratio	X
Group II (samples off three spools) [1/]	
Tensile proof	X
Mechanical strippability	X
Dynamic tensile strength	X
Group III (same three spools – 2 km each) [1/, 3/, 4/, 5/]	
Thermal shock [6/]	X
Storage temperature [7/]	X
Temperature humidity cycling [7/]	X
Temperature cycling [7/]	X
Temperature life (life aging) [6/, 8/]	X
Mechanical strippability (post temperature life)	X
Group IV	
Nuclear radiation resistance [7/]	X
Fungus resistance	X

NOTE: MM = Multimode

FOOTNOTES:

1/ See table I in MIL-PRF-49291 for specimen lengths.

2/ End view measurement acceptable for M49291/6-05BF.

3/ OOT = Change in optical transmittance. Perform in accordance with TIA-455-20 for transmitted power adhering strictly to the setup and test procedure specified in Measurement 2102 of MIL-STD-1678-2.

4/ Sample lengths: Recommend to prepare before group 1. For mechanical strippability, place in pans and remove from chamber after specified environmental test.

Mechanical strippability: each spool, 2 samples, 91.4 cm (3 ft) long.

Mechanical strippability - post temp/humidity: each spool, 2 samples, 91.4 cm (3 ft) long.

Mechanical strippability - post life aging: each spool, 2 samples, 91.4 cm (3 ft) long.

Dynamic tensile strength: each spool, 30 samples, 198.1 cm (6.5 ft) long (no humidity).

Nuclear radiation resistance: each spool, 1 sample, 200 meters (165.1 ft) long on spool.

Fungus resistance: each spool, 1 sample, 30.5 cm (1 ft) long.

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TABLE IV. Qualification inspection - Continued.

5/	Prepare in loose coils and place in pans. Recommend to terminate with single ferrule connectors (such as straight tip [ST] connectors) on the fiber ends (for attachment to test instrumentation optical ports – to obtain more stabilized measurements).
6/	Perform OOT after the test only.
7/	Perform OOT both during and after the test.
8/	Measurements requested daily for informational purposes.

In group A inspection testing, length may be measured using mechanical methods.

In group C inspection testing, the mechanical strippability test may be omitted if the optical fiber coatings have not changed from when the mechanical strippability test was last performed and the manufacturer provides a certificate of compliance for mechanical strippability.

The following additional requirements and tests are applicable to space qualified part or identifying numbers (PIN):

a. Storage temperature: Applicable.

b. Thermal vacuum outgassing: Applicable.

c. Odor: Applicable.

d. Toxicity: Applicable.

PINs (see figure 1 and table V):

M49291/6-02S

M49291/6-03

M49291/6-04S

M49291/6-05

M49291/6-05BF

TABLE V. Supersession data.

PIN	Superseding
M49291/6-02S	None
M49291/6-04S	None
M49291/6-03	D49291/01-006 1/
	M49291/6-01
	M49291/6-02
M49291/6-05	None
M49291/6-05BF	M49291/6-05 2/

NOTES:
1/ PIN is as shown in MIL-F-49291/6(NAVY).
2/ Intended for use in blown fiber applications with MIL-PRF-85045 tube cable.

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Qualification by similarity:

Manufacturers who are qualified under this specification sheet and whose optical fiber, with a change in the glass (composition, profile, etc.), passes the tests listed in table VI are qualified under this specification sheet for the optical fiber with changed glass.

TABLE VI. Qualification by similarity: change in glass.

Fiber length
Attenuation uniformity
Attenuation rate
Numerical aperture (MM only)
Core diameter (MM only)
Transient attenuation (MM only)
Macrobend attenuation
Bandwidth (MM only)
Chromatic dispersion
Temperature cycling
Nuclear radiation resistance

Manufacturers who are qualified under this specification sheet and whose optical fiber, with a change in the coating (composition, thickness, etc.), passes the tests listed in table VII are qualified under this specification sheet for the optical fiber with changed coating.

TABLE VII. Qualification by similarity: change in coating.

Fiber length
Attenuation rate
Transient attenuation (MM only)
Macrobend attenuation
Coating diameter
Overall coating concentricity ratio
Mechanical strippability
Dynamic tensile strength
Thermal shock
Storage temperature
Temperature humidity cycling
Temperature cycling
Temperature life (life aging)
Fungus resistance

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Manufacturers who are qualified under this specification sheet and whose optical fiber, with a change in the buffer build-up from 250 µm to 500 µm, passes the tests listed in table VIII are qualified under this specification sheet for the optical fiber with changed buffer build-up.

TABLE VIII. Qualification by similarity: change in buffer build-up.

Attenuation rate
Transient attenuation (MM only)
Mass/unit length
Coating diameter
Overall coating concentricity ratio
Mechanical strippability
Storage temperature
Temperature humidity cycling
Temperature cycling
Temperature life (life aging)
Fungus resistance

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC
NASA – NA

Preparing activity:
Navy – SH
(Project 6010-2018-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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